

QUESTION AND ANSWERS

1. Find the Fourier coefficients and Fourier series of the square-wave function f defined by

$$f(x) = \begin{cases} 0 & \text{if } -\pi \leq x < 0, \\ 1 & \text{if } 0 \leq x < \pi \end{cases} \quad \text{and } f(x + 2\pi) = f(x),$$

so f is periodic with period 2π .

Solution: Using the formulas for the Fourier coefficients in Definition, we have

$$a_0 = \frac{1}{2} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{2\pi} \int_{-\pi}^0 0 dx + \frac{1}{2\pi} \int_0^{\pi} 1 dx = 0 + \frac{1}{2\pi} (\pi) = \frac{1}{2}, \text{ and, for } n \geq 1$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_{-\pi}^0 0 dx + \frac{1}{\pi} \int_0^{\pi} \cos nx dx = 0$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_{-\pi}^0 0 dx + \frac{1}{\pi} \int_0^{\pi} \sin nx dx = \begin{cases} 0 & \text{if } n \text{ is even} \\ 2/n\pi & \text{if } n \text{ is odd.} \end{cases}$$

The Fourier series of f is therefore

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) = \frac{1}{2} + \frac{2}{\pi} \sin x + \frac{2}{3\pi} \sin 3x + \frac{2}{5\pi} \sin 5x + \dots$$

Since odd integers can be written as $n = 2k - 1$, where k is an integer, we can write the Fourier series in sigma notation as

$$\frac{1}{2} + \sum_{k=1}^{\infty} \frac{2}{(2k-1)\pi} \sin(2k-1)x$$

2. Find the Fourier series of $f(x) = \frac{\pi-x}{2}$ in $0 < x < 2\pi$ Hence deduce that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$.

Solution: The Fourier series of $f(x)$ having period 2π is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \dots \dots \dots (1)$$

$$\text{Where } a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx = \frac{1}{\pi} \int_0^{2\pi} \frac{\pi-x}{2} dx = 0$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_0^{2\pi} \frac{\pi-x}{2} \cos nx dx = 0 \quad (\text{apply Bernoulli's rule})$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_0^{2\pi} \frac{\pi-x}{2} \sin nx dx = \frac{1}{n} \quad (\text{again by Bernoulli's rule})$$

Thus, by substituting the values of a_0, a_n, b_n in (1) the Fourier series is given by

$$f(x) = \frac{\pi-x}{2} = \sum_{n=1}^{\infty} \frac{1}{n} \sin nx$$

To deduce the required series we put $x = \frac{\pi}{2}$ in the Fourier series of $f(x)$

$$f\left(\frac{\pi}{2}\right) = \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi}{2} \quad \text{on expanding we get,}$$

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

3. Find the Fourier series of $f(x) = x(2\pi - x)$ in $0 < x < 2\pi$

$$\text{Solution: } f(2\pi - x) = (2\pi - x)(2\pi - 2\pi - x) = x(2\pi - x) = f(x)$$

$f(x)$ is even in $(0, 2\pi)$ and hence $b_n = 0$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{2\pi} (2\pi x - x^2) dx = \frac{4\pi^2}{3}, \text{ on integration}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_0^{2\pi} (2\pi x - x^2) \cos nx dx = \frac{-4}{n^2}, \text{ on integration}$$

The required Fourier series is

$$f(x) = \frac{4\pi^2}{6} + \sum_{n=1}^{\infty} \frac{-4}{n^2} \cos nx$$

4. Obtain the sine half range Fourier series of $f(x) = x^2$ in $0 < x < \pi$

Solution: the sine half range Fourier series of the function $f(x)$ in $(0, \pi)$ is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx \text{ where } b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx$$

$$\text{We have, } b_n = \frac{2}{\pi} \int_0^{\pi} x^2 \sin nx dx \quad \text{applying Bernoulli's rule,}$$

$$b_n = \frac{2}{\pi} \left\{ \frac{(-1)^{n+1} \pi^2}{n} - \frac{2}{n^3} [1 - (-1)^n] \right\}$$

Thus, the required sine half range Fourier series is given by

$$f(x) = \sum_{n=1}^{\infty} \frac{2}{\pi} \left\{ \frac{(-1)^{n+1} \pi^2}{n} - \frac{2}{n^3} [1 - (-1)^n] \right\} \sin nx$$

5. Find the Fourier series of $f(x) = x(2\pi - x)$ in $0 < x < 2\pi$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l}$$

where

$$a_0 = \frac{2}{l} \int_0^l f(x) dx$$

$$= \int_0^1 x dx + \int_1^2 (2 - x) dx$$

$$= \left[\frac{x^2}{2} \right]_0^1 + \left[2x - \frac{x^2}{2} \right]_1^2$$

$$= 1$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx$$

$$= \frac{2}{2} \int_0^2 f(x) \cos \frac{n\pi x}{2} dx$$

$$= \int_0^1 x \cos \frac{n\pi x}{2} dx + \int_1^2 (2 - x) \cos \frac{n\pi x}{2} dx$$

$$= \left[x \cdot \frac{\sin \frac{n\pi x}{2}}{\frac{n\pi}{2}} \right]_0^1 - \left[-\frac{\cos \frac{n\pi x}{2}}{\frac{n^2 \pi^2}{4}} \right]_0^1 + \left[(2 - x) \cdot \frac{\sin \frac{n\pi x}{2}}{\frac{n\pi}{2}} \right]_1^2 - \left[\frac{\cos \frac{n\pi x}{2}}{\frac{n^2 \pi^2}{4}} \right]_1^2$$

$$= \frac{2}{n\pi} \sin \left(\frac{n\pi}{2} \right) + \frac{4}{n^2 \pi^2} \left[\cos \left(\frac{n\pi}{2} \right) - 1 \right] - \frac{2}{n\pi} \sin \left(\frac{n\pi}{2} \right) - \frac{4}{n^2 \pi^2} \left[\cos n\pi - \cos \left(\frac{n\pi}{2} \right) \right]$$

$$= \frac{8}{n^2 \pi^2} \cos \left(\frac{n\pi}{2} \right) - \frac{4}{n^2 \pi^2} [1 + (-1)^n]$$

b) The Half range Fourier Sine series is given by, $f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$

$$\text{where } b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx$$

$$\begin{aligned}
 &= \frac{2}{2} \int_0^2 f(x) \sin \frac{n\pi x}{2} dx \\
 &= \int_0^1 x \sin \frac{n\pi x}{2} dx + \int_1^2 (2-x) \sin \frac{n\pi x}{2} dx \\
 &= \left[x \cdot \frac{-\cos \frac{n\pi x}{2}}{\frac{n\pi}{2}} \right]_0^1 - \left[-\frac{\sin \frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}} \right]_0^1 + \left[(2-x) \cdot \frac{-\cos \frac{n\pi x}{2}}{\frac{n\pi}{2}} \right]_1^2 - \left[\frac{\sin \frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}} \right]_1^2 \\
 &= \frac{-2}{n\pi} \cos\left(\frac{n\pi}{2}\right) + \frac{4}{n^2\pi^2} \left[\sin\left(\frac{n\pi}{2}\right) \right] + \frac{2}{n\pi} \cos\left(\frac{n\pi}{2}\right) - \frac{4}{n^2\pi^2} \left[0 - \sin\left(\frac{n\pi}{2}\right) \right] \\
 &= \frac{8}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right)
 \end{aligned}$$

Therefore,

i. Half range Cosine series: $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{2}$

$$\Rightarrow f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{8}{n^2\pi^2} \cos\left(\frac{n\pi}{2}\right) - \frac{4}{n^2\pi^2} [1 + (-1)^n] \cos \frac{n\pi x}{2}$$

ii. Half range Sine series : $f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$

$$\Rightarrow f(x) = \sum_{n=1}^{\infty} \frac{8}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) \sin \frac{n\pi x}{2}$$